Lecture 10: LTI Resistive Circuits and Equivalent One-Ports

EE201 – Circuit Theory I

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current voltage a part of a circuit (box N)

Review: equations from the graph alone

	incidence / mesh	**fundamental cutset / loop**
KCL	$A\mathbf{i} = \mathbf{0}$	$Q\mathbf{i} = \mathbf{0}$
KVL	$M\mathbf{v} = \mathbf{0}$	$B\mathbf{v} = \mathbf{0}$
$n-1$ voltages	node voltages $\mathbf{e}$	tree voltages $\mathbf{v}_T$
$b-n+1$ currents	mesh currents $\mathbf{i}_M$	co-tree currents $\mathbf{i}_C$
works for	M: planar, nonseparable	any connected graph

KCL and KVL depend only on the graph: they hold for any elements.

A circuit has $2b$ unknowns: b more equations come from the elements.

Why this lecture?

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  - substitution theorem, equivalent one-ports
  - series and parallel connections, source transformation

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- **This lecture — transform:** replace a part by a source, or by a simpler part with the same terminal behaviour
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  - series and parallel connections, source transformation
- **Later — the circuit as it is:** node voltage and mesh current analysis

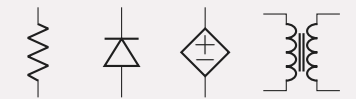
Roadmap

1. **LTI resistive circuits** — what the words mean, and what we gain
2. **A first example** — simplify a circuit step by step
3. **Substitution theorem** — replace a part of a circuit by a source
4. **One-ports and equivalence** — the same i–v characteristic at the port
5. **Series and parallel** — combine resistors and sources
6. **One source and one resistor** — exchange a voltage source and a current source

Resistive and dynamic elements

Resistive element: no memory

Its law relates v and i at the same instant, algebraically.

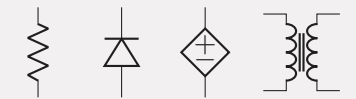


later: the ideal op-amp

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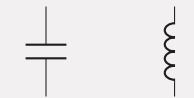
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Dynamic element: memory

Its law has a derivative.

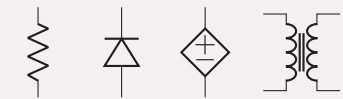


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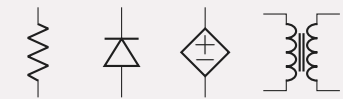
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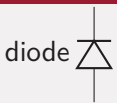
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"Resistive" means memoryless, not only "a resistor".

An element is linear if its law is linear in v, i; time-invariant if its law does not change with time.

Three questions about a circuit

Linear?

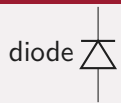


Nonlinear if it contains at least one nonlinear element, e.g. a diode.

Otherwise **linear**.

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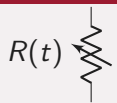
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Time-invariant?

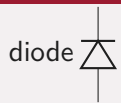


Time-varying if it contains at least one time-varying element, e.g. $R(t)$.

Otherwise **time-invariant**.

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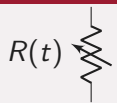
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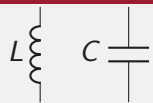
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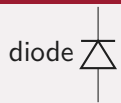


Dynamic if it contains at least one dynamic element: an L or a C.

Otherwise **resistive**.

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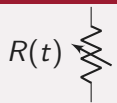
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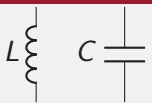
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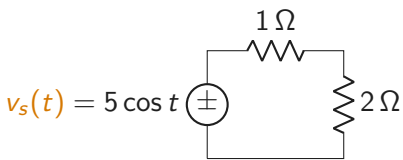
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LTI resistive circuit = linear + time-invariant + resistive:

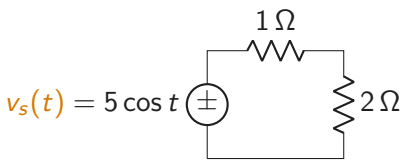
apart from the independent sources, every element is linear, time-invariant and resistive.

Independent sources are inputs



The source varies in time. Is the circuit time-varying?

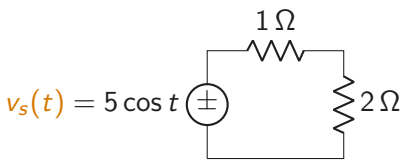
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The three questions are about the circuit that the inputs drive, not about the inputs. This is an LTI resistive circuit.

What varies in time?

In every circuit $v(t)$ and $i(t)$ are functions of time. The question is whether the element law changes with time.

Time-invariant resistor

$$v(t) = R\, i(t)$$

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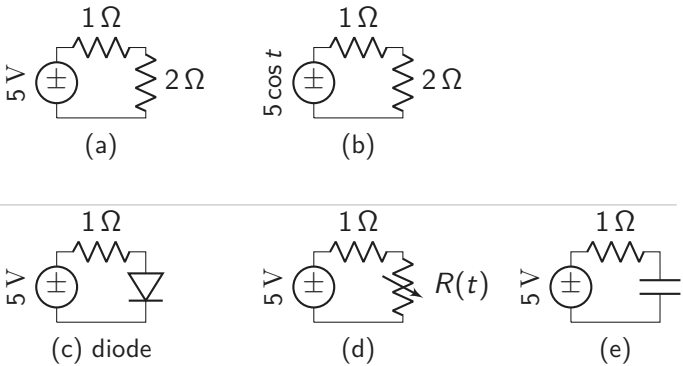
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Linear or nonlinear is also a property of the element law: $v = Ri$ is linear, a diode law is not.

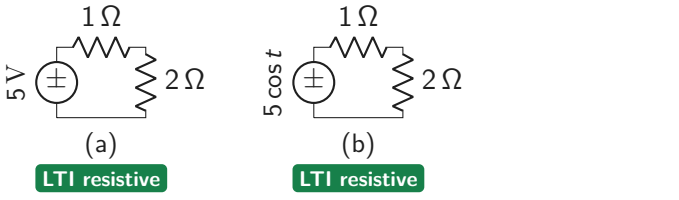
Which circuit is which?



A source, a $1\,\Omega$ resistor and one more element. In (d), $R(t) = 2 + \sin t\ \Omega$.

Linear? Time-invariant? Resistive?

Which circuit is which?



The input does not change the class: (a) and (b) are both LTI resistive.

Each of (c), (d), (e) breaks exactly one of the three properties.

What we gain

Always — no assumption on the elements

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Because the circuit is LTI resistive

Every element equation is linear and algebraic, with no derivative:

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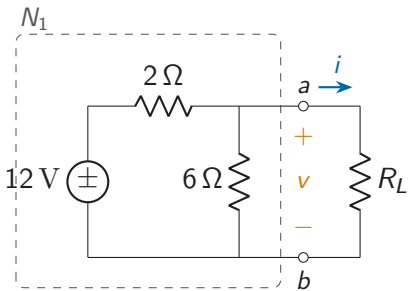
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- Time enters only through the values of the independent sources.
- So at each instant the circuit is solved like a circuit with constant sources: $2b$ linear algebraic equations.

Roadmap

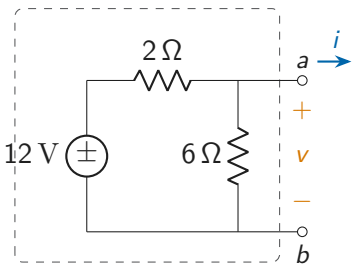
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A first example: what does R_L see?



R_L (the load) can be any resistive element.
Find v and i.

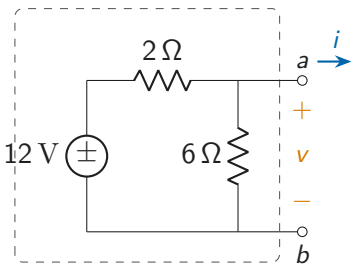
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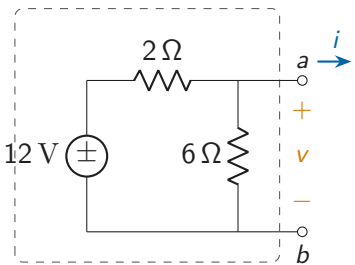


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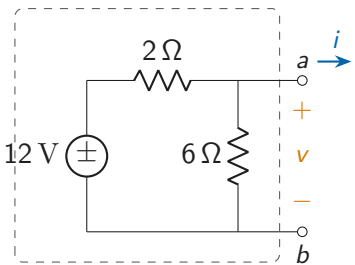
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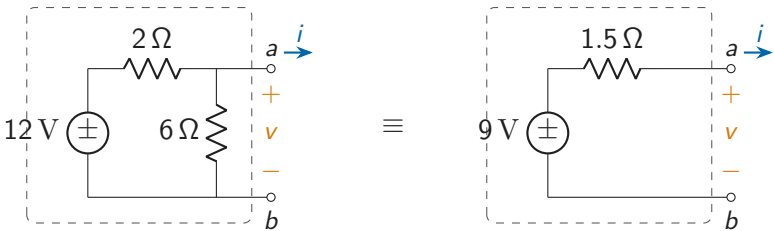
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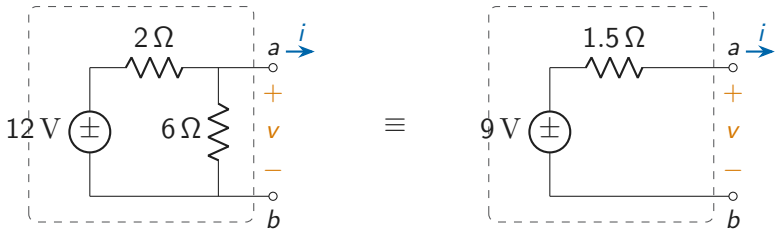
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The right one obeys $v = 9 - 1.5\,i$ too (KVL): the same law.

Read the result as a circuit

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Any load: $v = R_L i$, so $i = \dfrac{9}{1.5 + R_L}$. For $R_L = 3\,\Omega$: $i = 2\,\mathrm{A}$, $v = 6\,\mathrm{V}$.

What did we just do?

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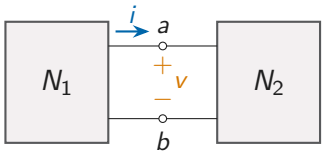
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question	answered by
May a part be replaced by a source?	substitution theorem
When may a part be replaced by a simpler part?	equivalent one-ports
Which simple replacements are there?	series, parallel, source transformation

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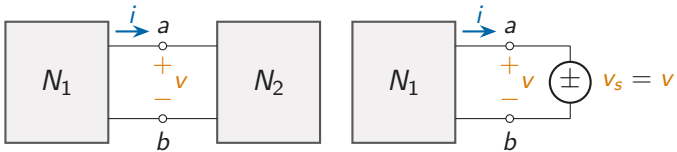
The substitution theorem



(I)

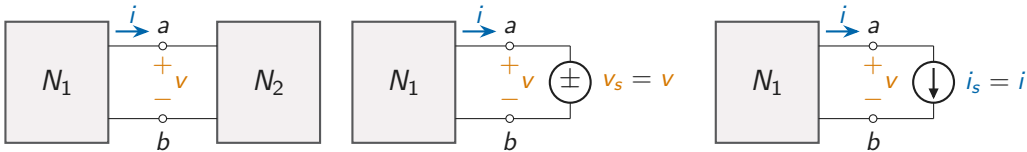
A circuit cut into two parts, joined at the terminals a, b.
i: current from N_1 into N_2 v: voltage between a and b

The substitution theorem



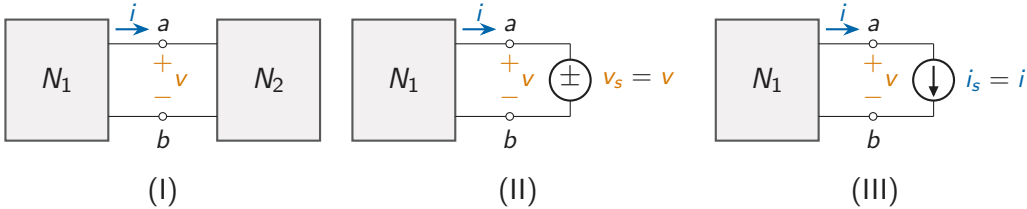
Replace N_2 by a voltage source with $v_s(t) = v(t)$ ...

The substitution theorem



... or by a current source with $i_s(t) = i(t)$.

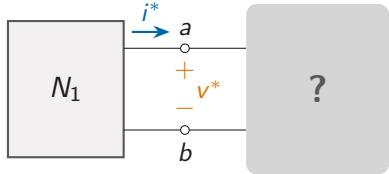
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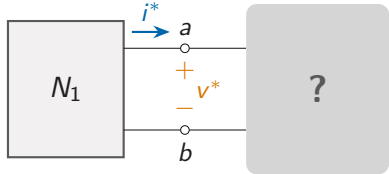
No branch voltage or branch current inside N_1 changes. An observer inside N_1 cannot tell (I), (II) and (III) apart.

Why does it work?



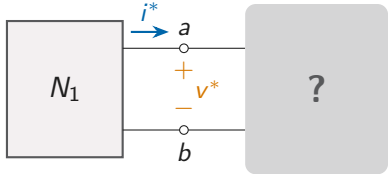
N_1 meets the rest of the circuit only at a and b.

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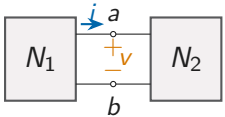


From inside N_1, the rest is just a voltage v^* and a current i^*.

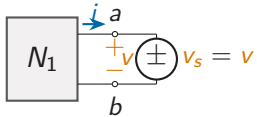
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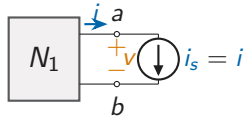
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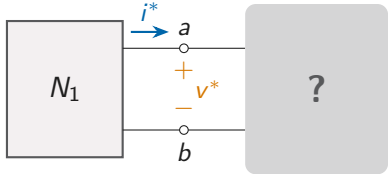


(II) v fixed at v^*, i free $\rightarrow i^*$

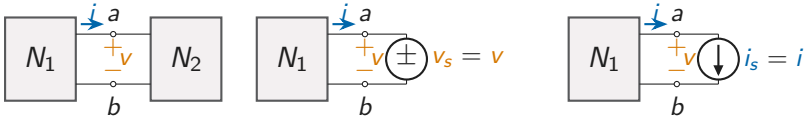


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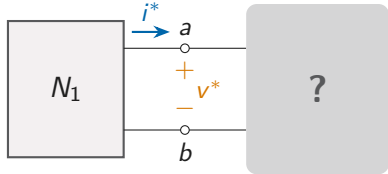
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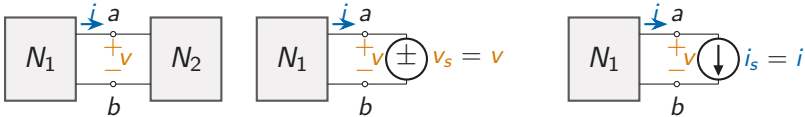
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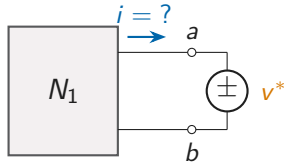


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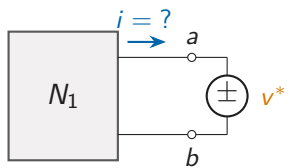
N_2 itself never enters: it can be anything, even nonlinear.

Why is the current i^* again?



Problem P: N_1 with a known voltage v^* at a–b.

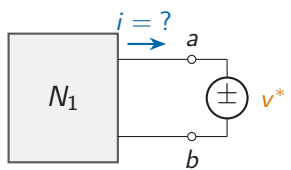
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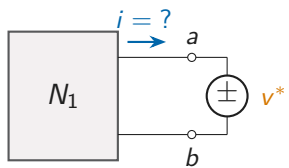
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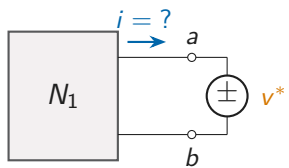
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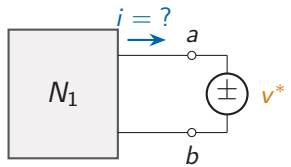
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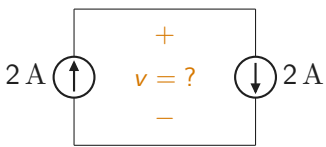
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In the first example

Fix $v = 6\,\mathrm{V}$. KCL at a: $\dfrac{12-6}{2} = \dfrac{6}{6} + i \;\Rightarrow\; i = 2\,\mathrm{A} = i^*$.

For a current source, P fixes $i = i^*$ instead, and $v = v^*$ follows the same way.

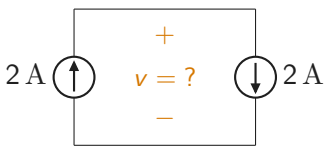
One condition: a unique solution



N_1 in place of N_2

N_1: a $2\,\mathrm{A}$ source. N_2 replaced by a $2\,\mathrm{A}$ source.

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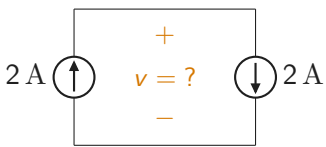


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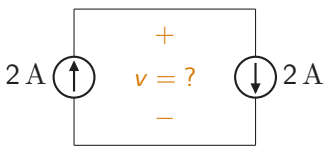
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One condition: a unique solution



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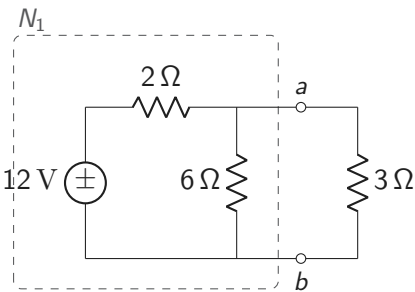
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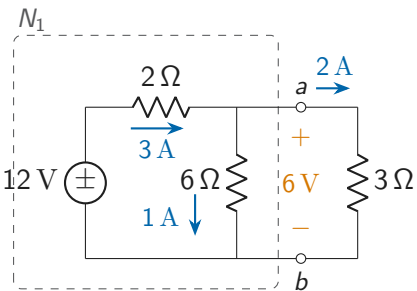
The substitution needs the new circuit to have a unique solution.

Example: substitution



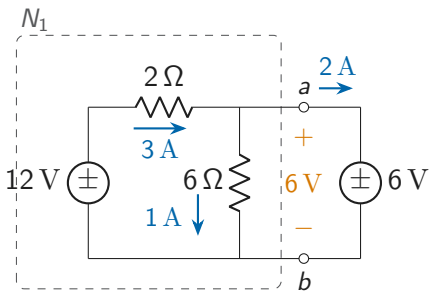
The first example with $R_L = 3\,\Omega$, cut at a, b: N_2 is the load.

Example: substitution



(I) From the first example ($R_L = 3\,\Omega$): $v = 6\,\mathrm{V}$, $i = 2\,\mathrm{A}$; inside N_1: $3\,\mathrm{A}$ in the $2\,\Omega$, $1\,\mathrm{A}$ in the $6\,\Omega$.

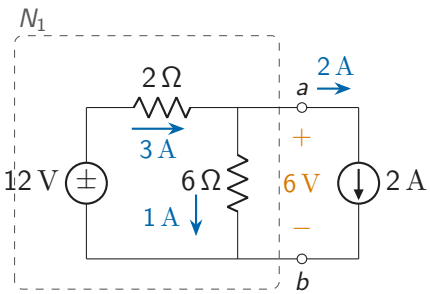
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(III) A $2\,\mathrm{A}$ source: KCL at a, $\frac{12-v}{2} = \frac{v}{6} + 2$, so $v = 6\,\mathrm{V}$: again $3\,\mathrm{A}$ and $1\,\mathrm{A}$. ✓

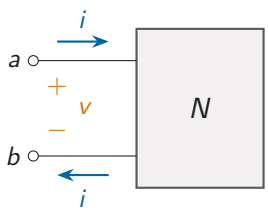
Roadmap

1. **LTI resistive circuits** — what the words mean, and what we gain ✓
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Ports and one-ports

Port, one-port

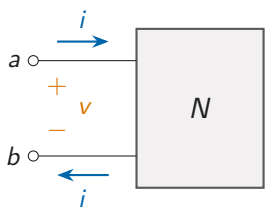
A pair of terminals is a port if the current entering one terminal equals the current leaving the other. A one-port is a circuit, or a circuit element, with one pair of terminals accessible from outside.



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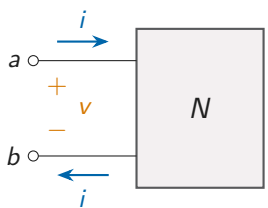


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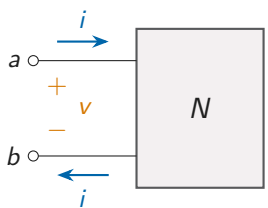


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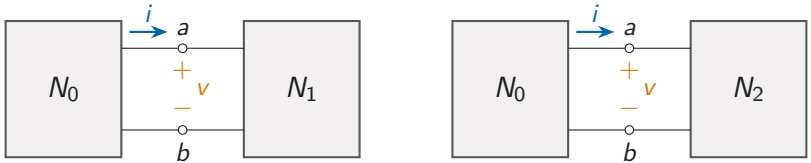


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- i enters the $+$ terminal of v.
- The relation between v and i is the i–v characteristic, e.g. $v = Ri$ for a resistor.

Equivalent one-ports

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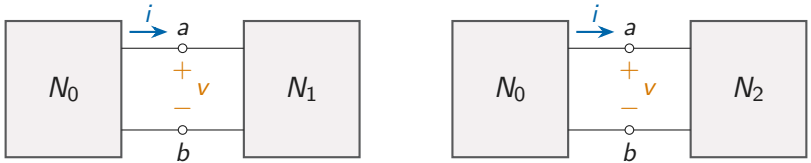
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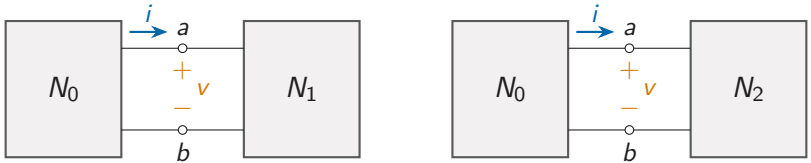


For any N_0: every branch voltage and current inside N_0 is the same in both.

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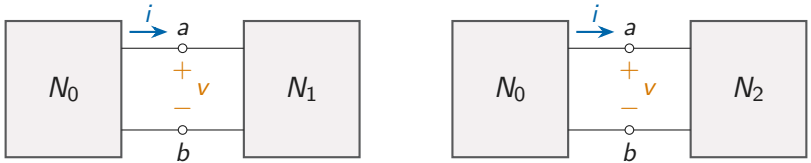
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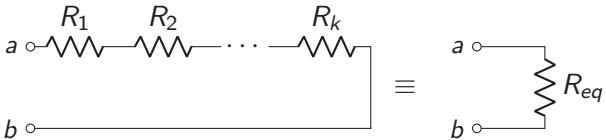
Inside N_1 and N_2 the branch values may differ.

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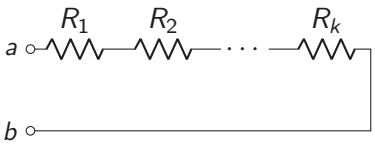
Series connection

Same current i through every element (KCL); the voltages add (KVL).

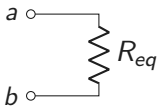


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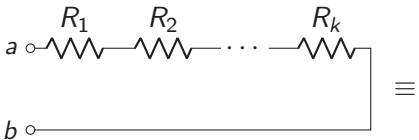


$$R_{eq} = R_1 + R_2 + \cdots + R_k$$

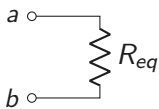
$$v = R_1 i + R_2 i + \cdots + R_k i = R_{eq} i$$

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$\equiv$

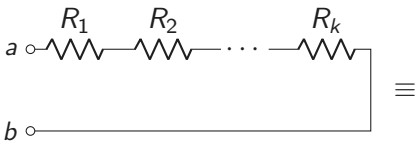


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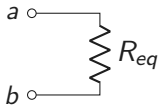
Voltage divider: the share of R_j is $v_j = \dfrac{R_j}{R_{eq}}\, v$

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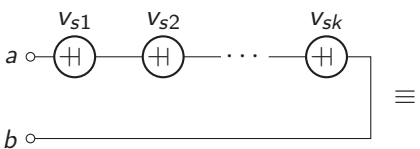
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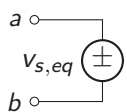
$\equiv$



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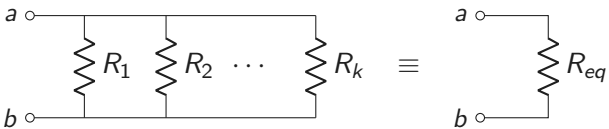


$$v_{s,eq} = v_{s1} + v_{s2} + \cdots + v_{sk}$$

All $+$ signs towards a; a source turned the other way enters with $-$.

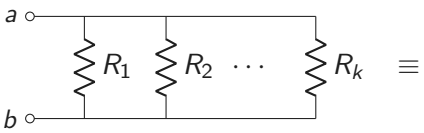
Parallel connection

Same voltage v across every element (KVL); the currents add (KCL).



Parallel connection

Same voltage v across every element (KVL); the currents add (KCL).

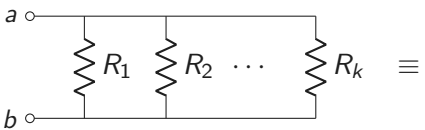


$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \cdots + \frac{1}{R_k}$$

With $G_j = 1/R_j$: $\quad i = G_1 v + \cdots + G_k v = G_{eq} v, \quad G_{eq} = G_1 + \cdots + G_k$

Parallel connection

Same voltage v across every element (KVL); the currents add (KCL).

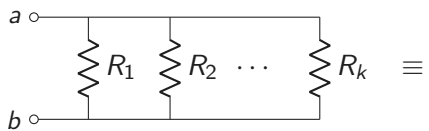


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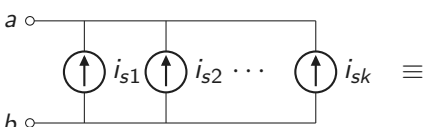
Current divider: $i_j = \dfrac{G_j}{G_{eq}}\, i.$ Two resistors: $R_{eq} = \dfrac{R_1 R_2}{R_1 + R_2}$

Parallel connection

Same voltage v across every element (KVL); the currents add (KCL).



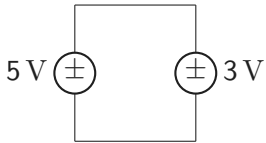
$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \cdots + \frac{1}{R_k}$$



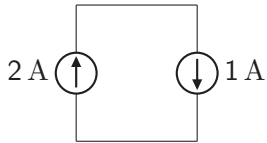
$$i_{s,eq} = i_{s1} + i_{s2} + \cdots + i_{sk}$$

All arrows towards a; a source turned the other way enters with $-$.

Two connections to avoid



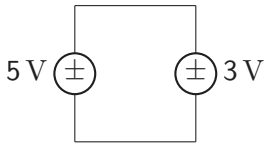
two voltage sources in parallel



two current sources in series

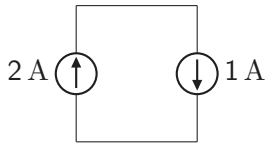
Are these allowed?

Two connections to avoid



two voltage sources in parallel

KVL fails: $5 \neq 3$.



two current sources in series

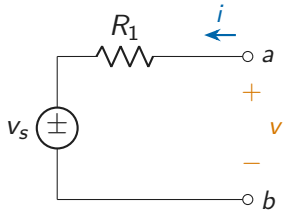
KCL fails: $2 \neq 1$.

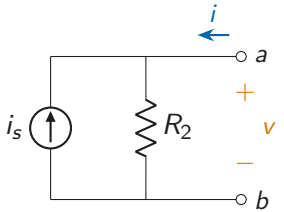
With equal values they are allowed, but the shared current (or voltage) is not fixed.

Roadmap

1. **LTI resistive circuits** — what the words mean, and what we gain ✓
2. **A first example** — simplify a circuit step by step ✓
3. **Substitution theorem** — replace a part of a circuit by a source ✓
4. **One-ports and equivalence** — the same i–v characteristic at the port ✓
5. **Series and parallel** — combine resistors and sources ✓
6. **One source and one resistor** — exchange a voltage source and a current source

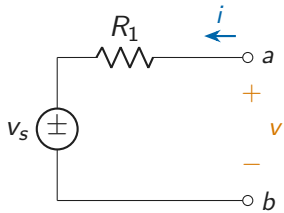
One source and one resistor

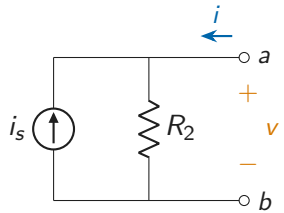




When do these two one-ports have the same i–v characteristic?

One source and one resistor

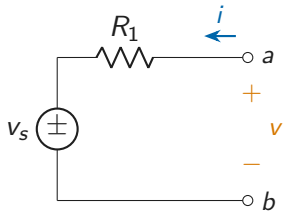




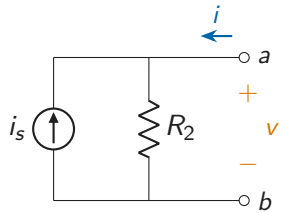
$$v = R_1 i + v_s$$

KVL around the loop: R_1 carries i.

One source and one resistor



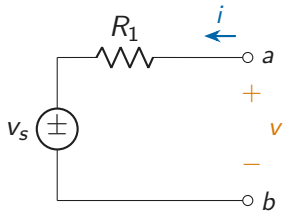
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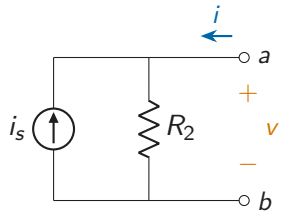
$$v = R_2(i + i_s) = R_2 i + R_2 i_s$$

i and i_s both flow into the top node: R_2 carries $i + i_s$.

One source and one resistor



$$v = R_1 i + v_s$$



$$v = R_2(i + i_s) = R_2 i + R_2 i_s$$

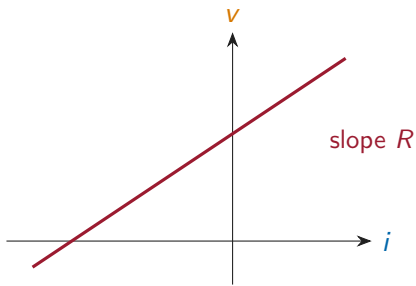
Both are straight lines in the i–v plane: equal slopes and equal intercepts.

Source transformation

Source transformation

The two one-ports are equivalent if

$$R_1 = R_2 = R \qquad \text{and} \qquad v_s = R\, i_s.$$



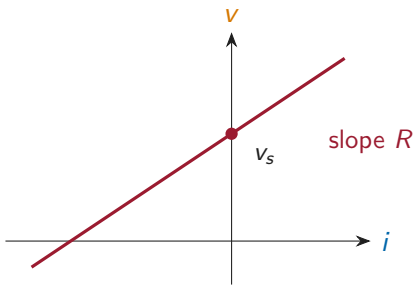
Source transformation

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Open circuit ($i = 0$): $v = v_s$.



Source transformation

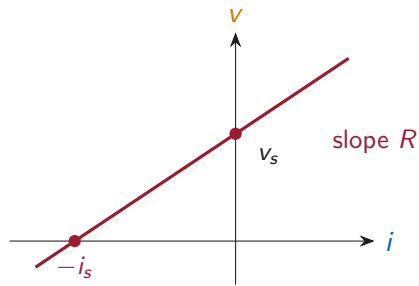
Source transformation

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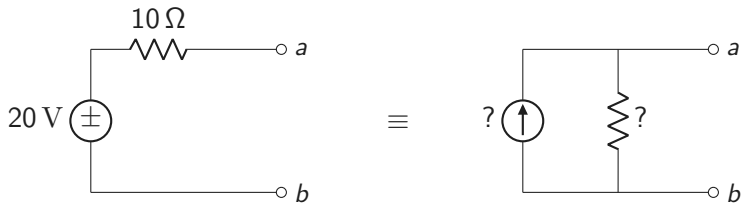
$$R_1 = R_2 = R \qquad \text{and} \qquad v_s = R\, i_s.$$

Open circuit ($i = 0$): $v = v_s$.

Short circuit ($v = 0$): $i = -i_s$.

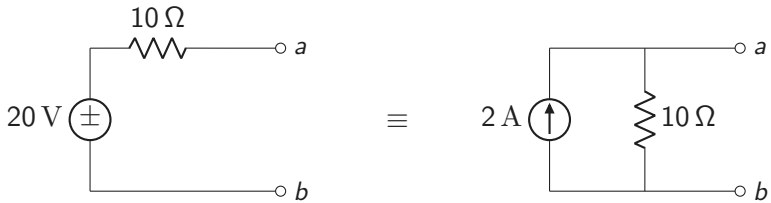


Example: a source transformation



Find the current source and the resistor.

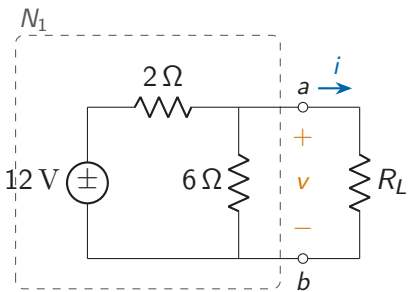
Example: a source transformation



$$R = 10\,\Omega, \qquad i_s = v_s/R = 20/10 = 2\,\mathrm{A}$$

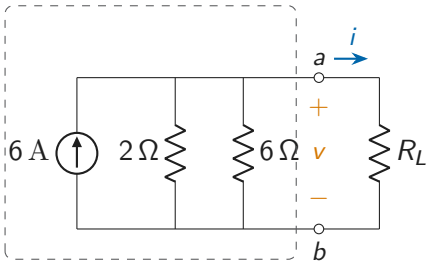
Both: open-circuit voltage $20\,\mathrm{V}$, short-circuit current $2\,\mathrm{A}$ (out of a).

The first example, again



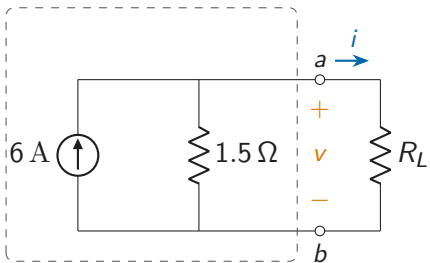
Now simplify only the part left of a–b; R_L stays.

The first example, again



Step 1. Source transformation:
$12\,\mathrm{V}$ with $2\,\Omega \rightarrow 6\,\mathrm{A}$ with $2\,\Omega$ $(12 = 2 \times 6)$

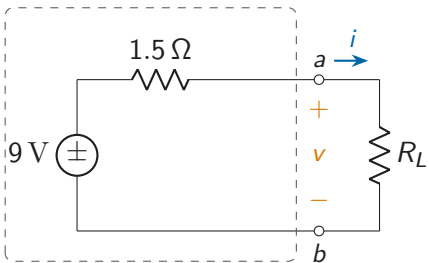
The first example, again



Step 1. Source transformation:
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Step 2. Parallel: $2\,\Omega \parallel 6\,\Omega = 1.5\,\Omega$

The first example, again

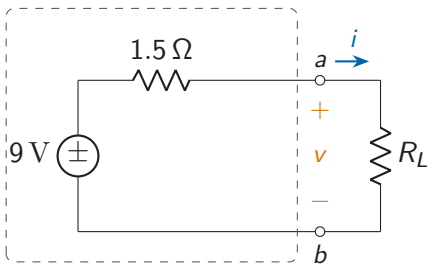


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Step 3. Source transformation:
$6\,\mathrm{A}$ with $1.5\,\Omega \rightarrow 9\,\mathrm{V}$ with $1.5\,\Omega$
$(9 = 1.5 \times 6)$

The first example, again



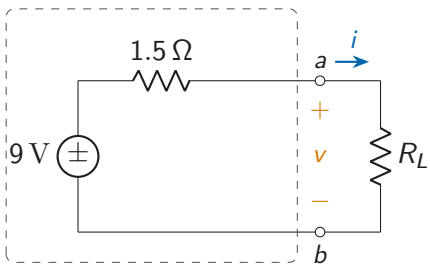
Step 1. Source transformation:
$12\,\mathrm{V}$ with $2\,\Omega \to 6\,\mathrm{A}$ with $2\,\Omega$ $(12 = 2 \times 6)$

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Step 3. Source transformation:
$6\,\mathrm{A}$ with $1.5\,\Omega \to 9\,\mathrm{V}$ with $1.5\,\Omega$
$(9 = 1.5 \times 6)$

Same $9\,\mathrm{V}$ and $1.5\,\Omega$, now by transforming the circuit.

The first example, again



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Step 3. Source transformation:
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$(9 = 1.5 \times 6)$

Same $9\,\mathrm{V}$ and $1.5\,\Omega$, now by transforming the circuit.

Careful: inside, the values differ; for $R_L = 3\,\Omega$ the $12\,\mathrm{V}$ source carries $3\,\mathrm{A}$, the $9\,\mathrm{V}$ source $2\,\mathrm{A}$.

Solving LTI resistive circuits

- Series and parallel connections and source transformations simplify many circuits, but not all.

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- Two methods write the equations of the circuit as it is, without transforming it, and work for every LTI resistive circuit:
  1. node voltage analysis: the $n-1$ node voltages are the unknowns;
  2. mesh current analysis: the $b-n+1$ mesh currents are the unknowns.

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  1. node voltage analysis: the $n-1$ node voltages are the unknowns;
  2. mesh current analysis: the $b-n+1$ mesh currents are the unknowns.
- Both start from the matrices met earlier (A, M), now with linear element equations.

Summary

- LTI resistive: apart from the independent sources (external inputs), every element is linear, time-invariant and memoryless, so all equations are linear and algebraic.
- Substitution: replace N_2 by a voltage source $v_s = v$ or a current source $i_s = i$; nothing changes inside N_1.
- Equivalent one-ports: the same i–v characteristic; the rest cannot tell them apart.
- Series: $R_{eq} = \sum R_j$, $v_{s,eq} = \sum v_{sj}$. Parallel: $G_{eq} = \sum G_j$, $i_{s,eq} = \sum i_{sj}$.
- Source transformation: v_s with series $R \equiv i_s = v_s/R$ with parallel R.